Wall time per step: Splat simulation, 500k (482,409 cells)

Seconds per step. Colour is scaled per step and size: fastest framework 0, slowest 1, on square−root seconds.

	bixverse mtx	bixverse h5ad	bixverse GPU*	bixverse MLX*	Scanpy mtx	Scanpy h5ad	Scanpy dask	scrapper mtx	scrapper h5ad	scrapper HDF5	Seurat std mtx	Seurat std h5ad	Seurat BPCells
load	48	29	21	19	OOM	11	21	205	23	224	225	41	66
qc	0.3	0.2	0.3	0.3	–	2.8	3.8	2.2	2.2	6.3	14	13	4.0
normalise	n/a	n/a	n/a	n/a	–	3.6	0.0	0.4	0.4	0.0	0.9	0.8	1.9
hvg	1.0	1.1	0.8	0.7	–	3.1	4.5	0.5	0.5	19	19	19	13
pca	2.3	2.3	1.3	1.1	–	10	11	14	14	31	OOM	OOM	271
harmony	1.7	1.6	1.1	0.7	–	2.1	2.0	100	84	82	–	–	88
knn_snn	14	14	7.5	6.7	–	33	29	40	40	40	–	–	45
cluster	4.1	4.0	3.2	3.1	–	20	20	116	110	102	–	–	132
cluster_fast	14	14	3.1	2.5	n/a	n/a	n/a	196	194	195	n/a	n/a	n/a
umap	35	34	3.9	3.7	–	226	226	213	219	213	–	–	256
dge	2.3	2.3	1.8	1.9	–	454	–	22	22	1,710	–	–	404
tsne_default	252	251	223	225	–	1,315	1,319	n/a	n/a	n/a	–	–	1,404
tsne_fft	63	61	14	20	–	153	156	n/a	n/a	n/a	–	–	157
tsne_quick	42	48	27	23	n/a	n/a	n/a	155	152	153	n/a	n/a	n/a
total (no tSNE)	109	88	41	37	failed	766	317	712	515	2,426	failed	failed	1,281

Black: n/a (step not in that framework), OOM or TIMEOUT where the run died, − not reached.
Each step is scaled over the runs that finished it: at 1M the bixverse CPU tSNE is red because the others timed out.
Scanpy dask has no DGE step. bixverse GPU and MLX run natively without a memory cap.
bixverse CPU tSNE builds its kNN with IVF from 500k (kmknn at 100k). (qd) steps are not in the total.